

**GSM BSS
GBSS22.1**

Half-Rate Service Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 About This Document

1.1 Scope

This document describes GBFD-113401 Half Rate Speech, including its technical principles and engineering guidelines.

1.2 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

The product supports only the functions mentioned in this document, and the functions can be enabled only in the specified scenarios. If a function not mentioned in this document is enabled or a function is enabled in an unspecified scenario, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

1.3 Change History

This section provides information about the changes in different document versions. There are two types of changes, which are defined as follows:

- Feature change
Changes in features of a specific product version
- Editorial change
Changes in wording or addition of information that was not described in the earlier version

01 (2020-04-07)

This issue does not include any changes.

Draft A (2020-01-18)

Compared with Issue 01 (2016-08-31) of GBSS18.1, Draft A (2020-01-18) of GBSS22.1 includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	Changed the name of CME to MAE-Deployment.	None
Editorial change	None	None

2 Overview

2.1 Introduction

With an increasing number of subscribers, GSM spectrum resources become insufficient. The Half Rate Speech feature increases the number of configurable voice channels. Therefore, it greatly improves the spectrum efficiency, doubles the network capacity, and saves the transmission resources on the Um interface without affecting the QoS greatly.

If the full-rate speech codec is used on the Um, Abis, and Ater interfaces, the BSS uses channels of 16 kbit/s for coding and transmission. If the half-rate speech codec is used on the Um, Abis, and Ater interfaces, the BSS uses channels of 8 kbit/s for coding and transmission. Therefore, when half-rate voice channels are used, a 16-kbit/s channel is used for the terrestrial circuit over the Abis interface to carry two voice services. This way, the terrestrial link can carry more traffic and therefore saves the transmission resources over the Abis interface.

According to 3GPP TS 46.002, to enable the BSS to support the half-rate speech codec, some modifications are required. The modifications involve the Um interface, Abis interface, Ater interface, physical channels, and transcoder and rate adapter unit (TRAU).

NOTE

The BSC6910 does not support the Ater interface.

The version of half-rate speech codec is Half-Rate Speech Version 1.

2.2 NEs Supporting the Feature

Table 2-1 NEs supporting the feature

Feature	BSC6900	BSC6910	GBTS	eGBTS
Half Rate Speech	√	√	√	√

 NOTE

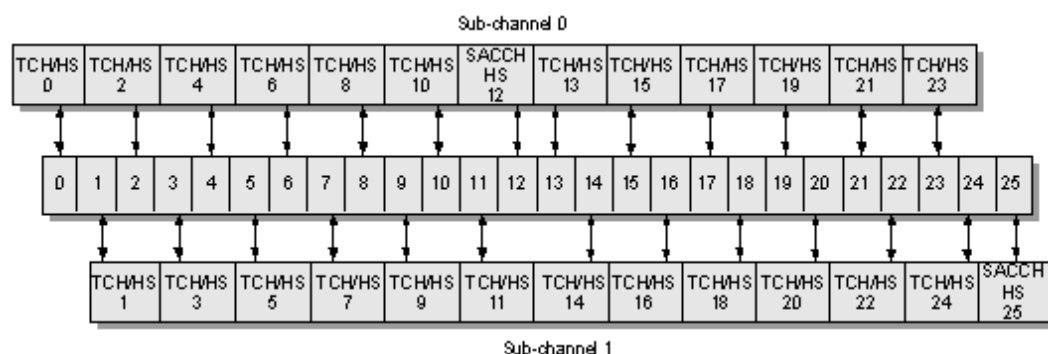
√ indicates that the NE supports this feature. × indicates that the NE does not support this feature.

3 Technical Description

3.1 Principles

To support Half Rate Speech, both the MS and the BTS must support GSM Phase 2. **Figure 3-1** shows the implementation of Half Rate Speech over the Um interface.

Figure 3-1 Implementation of Half Rate Speech over the Um interface



The MS performs the following procedures on the uplink and downlink when Half Rate Speech is enabled:

- On the downlink
The MS decodes only the odd- or even-numbered multiframes of the four multiframes. This means that the MS decodes only two multiframes out of four multiframes.
- On the uplink
The MS encodes only the odd- or even-numbered multiframes of the four multiframes. This means that the MS encodes only two multiframes out of four multiframes.

The BTS performs the following procedures on the uplink and downlink when Half Rate Speech is enabled:

- On the downlink
The BTS encodes the data on sub-channels 0 and 1.
 - For the two multiframes generated from the encoding of the data on sub-channel 0, the BTS transmits them through the even-numbered multiframes.
 - For the two multiframes generated from the encoding of the data on sub-channel 1, the BTS transmits them through the odd-numbered multiframes.

On the Um interface, the modulation scheme for half-rate speech signals is the same as that for full-rate speech signals.

- On the uplink
The BTS does not distinguish between half-rate speech and full-rate speech during multiframe demodulation over the Um interface. The BTS distinguishes between odd-numbered multiframes and even-numbered multiframes during multiframe decoding. After the decoding, the BTS puts the decoded multiframes onto sub-channels 0 and 1 separately.

NOTE

The **VOICEVER (BSC6910, BSC6900)** parameter specifies whether to enable Half-Rate Service.

Figure 3-1 shows sub-channels 0 and 1.

3.2 Channel Allocation on the Um Interface

The BSC allocates full-rate or half-rate channels based on the service type, service requirement, and cell load. For details, see *Channel Management Feature Parameter Description*. During channel allocation, the BSC performs conversion between full-rate and half-rate channels based on the cell load and channel type request. For details, see *Dynamic Channel Conversion Feature Parameter Description*.

3.3 Channel Allocation on the Abis Interface

The BSC determines whether to adjust the proportions of full-rate channels and half-rate channels based on the load of the transmission resources over the Abis interface. Therefore, when the resources are insufficient, the BSC can convert some full-rate channels to half-rate channels in time. For details, see *Dynamic Channel Conversion Feature Parameter Description*.

3.4 TRAU

To enable Half Rate speech on a BSC, the TRAU is required.

Huawei BSS allows multiple TRAUs to form a TC resource pool. The TC resource pool contains the TRAUs and transmission resources over the Ater interface and the A interface. These resources are shared, without fixed mapping between them. The TRAU supports the coding and decoding of full-rate speech, half-rate speech, enhanced full-rate speech, and AMR speech. During a call, a coding and decoding resource is selected from the resource pool.

 NOTE

The BSC6910 does not support the Ater interface.

3.5 AMR HR

The half-rate speech codec and the AMR HR speech codec have the same spectrum efficiency. The AMR HR speech codec, however, is more flexible and robust in terms of encoding rate.

For the differences between the codecs with regard to power control and handover, see *Power Control Feature Parameter Description* and *Handover Feature Parameter Description*.

3.6 DTX

Discontinuous Transmission (DTX) is an effective method of prolonging MS battery life and reducing radio interference. The use of Half Rate Speech does not affect the DTX. The DTX for half-rate speech is controlled by both the **HRDLDTX (BSC6900, BSC6910)** and **HRULDTX (BSC6900, BSC6910)** parameters. For details about the DTX, see *Discontinuous Transmission and Discontinuous Reception Feature Parameter Description*.

3.7 Handover

Handover algorithm includes the decision about AMR or non-AMR speech handovers between half-rate channels and full-rate channels. For details, see *Handover Feature Parameter Description*.

3.8 Power Control

To support Half Rate Speech, the power control algorithm is adjusted accordingly.

Huawei III power control algorithm uses different quality thresholds for different rates. For example, when a call is processed on a half-rate channel, the algorithm uses the half-rate quality threshold. When the call is handed over to a full-rate channel, the algorithm uses the full-rate quality threshold. For details, see *Power Control Feature Parameter Description*.

3.9 Local Switching

During a call, one party may use the full-rate speech codec while the other party may use the half-rate speech codec. In such a case, the BSC first changes the half-rate speech codec to a full-rate speech codec and then performs local switching. For details, see *BSS Local Switching Feature Parameter Description*.

3.10 Impact on KPIs

The impact of Half Rate Speech on KPIs is as follows:

- If full-rate channels are changed to half-rate channels, the number of occupied timeslots decreases and therefore the interference to the entire network decreases.
- If Half Rate Speech is enabled when the carrier-interface ratio (CIR) of the entire network is more than 10 dB, the speech quality on the entire network deteriorates. The higher the proportion of half-rate traffic on the entire network, the more sharply the speech quality on the entire network degrades.
- If Half Rate Speech is enabled when the CIR of the entire network is less than 10 dB, the speech quality on the entire network improves. The higher the proportion of half-rate traffic on the entire network, the more the speech quality of the entire network improves.
- The enabling of Half Rate Speech has a minor impact on relevant KPIs.

3.11 Impact on Equipment

For a BSC, half-rate speech requires higher processing capability than full-rate speech.

In addition, the use of Half Rate Speech increases the system capacity, and the CPU usage of the TC in the BSC increases slightly.

4 Engineering Guidelines

Multiplexing Ratio of the LAPD Link

If the half-rate mode is enabled in a cell, the recommended multiplexing ratio of the carrier's LAPD link is 2:1, that is, the bandwidth of the LAPD link is 32 kbit/s.

Number of Carriers Supported by Each E1 Link

The number of carriers supported by each E1 link on the Abis interface depends on the network configuration, as described in the following list:

- In the case that BTS local switching and Flex Abis are not enabled on the network
 - If the half-rate mode is enabled for all the carriers on an E1 link, a maximum of 13 carriers can be supported by the E1 link.
 - If both half-rate mode and full-rate mode are configured for the carriers on an E1 link, a maximum of 13 to 15 carriers can be supported by the E1 link.
- In the case that BTS local switching and Flex Abis are enabled on the network
The number of carriers supported by an E1 link depends on the proportion of calls processed through BTS local switching to all calls. Generally, a maximum of 13 to 18 carriers can be supported by an E1 link.
- In the case of transmission optimization
 - If the half-rate mode is enabled for all the carriers on an E1 link, a maximum of 20 carriers can be supported by the E1 link.
 - If both half-rate mode and full-rate mode are configured for the carriers on an E1 link, a maximum of 20 to 24 TRXs can be supported by the E1 link.

4.1 Deployment Requirements

Table 4-1 Deployment requirements

Aspect	Requirement
Related features	None
BSC	None
BTS	None
GSM networking	None
MS	MSs support the Half-Rate Service feature.
MSC	The MSC supports the Half-Rate Service feature.
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> .
Others	None

4.2 Activation

Using MML Commands

Step 1 On the BSC LMT, run the **SET GCELLCCACCESS** command with **Speech Version** set to **FULL_RATE_VER1(Full-rate VER 1)**, and **HALF_RATE_VER1(Half-rate VER 1)**. In the speech versions, **HALF_RATE_VER2(Half-rate VER 2)** is reserved and is unavailable to any service.

Step 2 (Optional) Enable the Dynamic Channel Conversion feature. For details, see *Dynamic Channel Conversion Feature Parameter Description*.

----End

MML Command Examples

```
SET GCELLCCACCESS: IDTYPE=BYID, CELLID=0,
VOICEVER=FULL_RATE_VER1-1&FULL_RATE_VER2-0&FULL_RATE_VER3-0&HALF_R
ATE_VER1-1&HALF_RATE_VER2-0&HALF_RATE_VER3-0&FULL_RATE_VER5-0;
```

Using the MAE-Deployment

NOTE

When configuring the Half Rate Speech feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required.

You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment. (MAE-Deployment single configuration)

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 4-2](#). (Optional) Modify objects in batches on the MAE-Deployment. (MAE-Deployment batch modification center)

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Batch Modification on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 4-2 Configuring the parameter on the MAE-Deployment

SN	Managed Object (MO)	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLCCACCESS	BSC6900/BSC6910	Speech Version	VOICEVER	Yes

4.3 Activation Observation

On the BSC LMT, run the **LST GCELLCCACCESS** command to query the value of **Speech Version**.

Expected result: The value of **Speech Version** is **ON**.

4.4 Deactivation

The Half Rate Speech feature cannot be deactivated because a BSC must be configured with a speech version. However, you can run the **SET GCELLCCACCESS** command to configure a required speech version.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

Base station controller parameter reference

- BSC6900 GSM Parameter Reference: contains the parameters related to the BSC6900 equipment, transport, and radio access functions.
- BSC6910 GSM Parameter Reference: contains the parameters related to the BSC6910 equipment, transport, and radio access functions.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the "Operation and Maintenance > Configuration Management" node in the *BSC69xx Product Documentation* delivered with that version.

eGBTS Parameter reference

- Node Parameter Reference: contains base station equipment and transport parameters.
- GBTSFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

1. Open the EXCEL file of parameter reference.
2. On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, GBFD-200201.
3. Click **OK**. All parameters related to the feature are displayed.

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

1. Open the EXCEL file of the used reserved parameter list.
2. On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

Base station controller performance counter reference

- BSC6900 GSM Performance Counter Reference: contains the counters related to the BSC6900 equipment, transport, and radio access functions.
- BSC6910 GSM Performance Counter Reference: contains the counters related to the BSC6910 equipment, transport, and radio access functions.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the "Operation and Maintenance > Performance Management" node in the *BSC69xx GSM Product Documentation* delivered with that version.

eGBTS performance counter reference

- Node Performance Counter Summary: contains base station equipment and transport counters.
- GBTSFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

1. Open the EXCEL file of performance counter reference.
2. On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, GBFD-200201.
3. Click **OK**. All counters related to the feature are displayed.

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see the *Glossary*.

8 Reference Documents

1. 3GPP TS 46.002 "Half rate speech; Half rate speech processing functions"
2. 3GPP TS 46.006 "Half rate speech; ANSI-C code for GSM half-rate speech codec"
3. 3GPP TS 46.007 "Half rate speech; Test sequences for the GSM half rate speech codec"
4. 3GPP TS 46.008 "Half rate speech; Performance Characterization of the GSM Half Rate speech codec"
5. 3GPP TS 46.020 "Half rate speech; Half rate speech transcoding"
6. 3GPP TS 46.021 "Half rate speech; Substitution and muting of lost frames for half rate speech traffic channels"
7. 3GPP TS 46.022 "Half rate speech; Comfort noise aspects for the half rate speech traffic channels"
8. 3GPP TS 46.041 "Half rate speech; Discontinuous Transmission (DTX) for half rate speech traffic channels"
9. 3GPP TS 46.042 "Half rate speech; Voice Activity Detector (VAD) for half rate speech traffic channels"
10. 3GPP TR 46.085 "Subjective tests on the interoperability of the Half Rate / Full Rate / Enhanced Full Rate (HR/FR/EFR) speech codecs, single, tandem and tandem free operation"
11. 3GPP TS 46.001 "Full rate speech; Processing functions"